

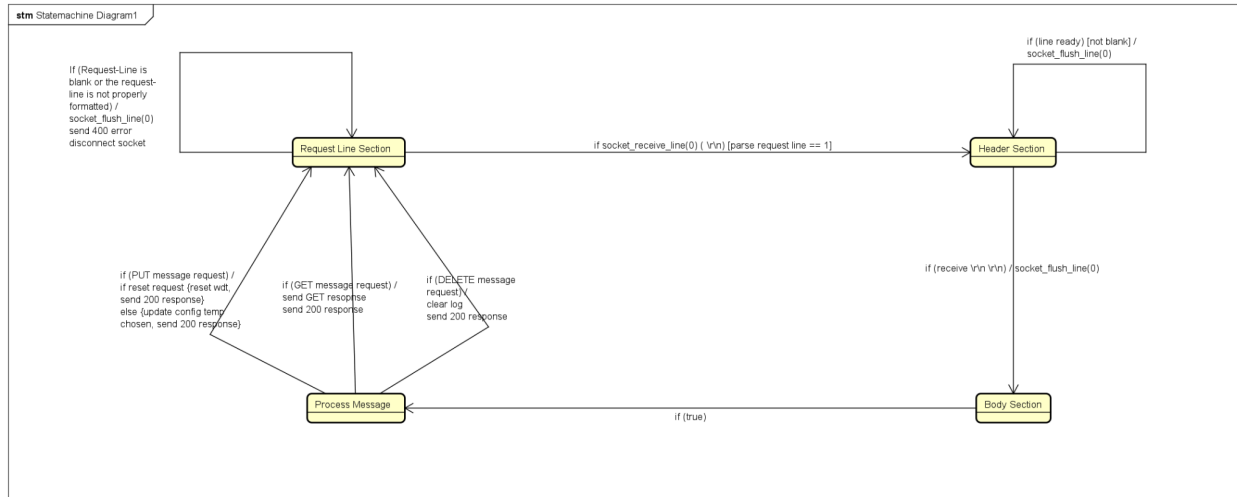


Figure 1. HTTP Parser FSM - The diagram illustrates a UML state machine for parsing incoming HTTP requests. The FSM contains four states: Request Line Section, Header Section, Body Section, and Process Message. The FSM starts in the Request Line Section and waits for a complete line in the socket receive buffer. A blank or malformed request line causes the FSM to flush the buffer, send a 400 error, disconnect, and remain in the Request Line Section. A valid request line advances the FSM to the Header Section. The Header Section discards incoming header lines until a blank line signals the end of the HTTP header block, at which point the FSM advances to the Body Section. The Body Section passes through immediately to Process Message, as all supported endpoints use query string parameters rather than a request body. In Process Message, the FSM dispatches based on the parsed method and URI, sends the appropriate response, disconnects the socket, and resets to the Request Line Section to await the next request.

<https://youtu.be/sew-Gzpn7f8>